NUMERICAL COMPANION

qEDA computations: what is calculated, where, and what it shows

A plain-language guide to the algebraic controls, matched topology test, Bowles benchmark families, Iris audit, and credit-card fraud case study.

The central question
After a table is embedded into a Hilbert space, does the resulting operator reproduce a declared classical view, add structure irrelevant to the analyst's question, or supply a stable additional description?

Audit	Data	Main answer
E1 controls	Synthetic states	The implementation satisfies the exact operator and phase identities.
E2 topology	Ring vs filled disk	Quantum-looking structure appears, but it does not detect the support hole.
E3 benchmark	55 Bowles variants	Spectral response is not a simple relabelling of the selected covariance summaries.
E4 Iris	150 native rows	Rows missed by classical probes have lower held-out class-subspace fidelity.
E4 fraud	284,807 released rows	qEDA ranks anomalies well; partial J adds almost no ranking information beyond J = 0.

Companion to Beyond the kernel: exploratory analysis of quantum encodings of tabular data through the class density operator.
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1. The common computation

Each row x is first mapped to a normalized state |psi(x,J)>. The rows are not compressed by PCA to satisfy a qubit budget. Standardisation is used to estimate dependence, while a separate train-fitted affine map places native features in the angle interval [0, pi].

```
input table -> fitted scaling -> angles x
x -> RY(x/2) -> ZZ(J) -> RY(x/2) -> |psi(x,J)>
class rows -> rho_c = (1/M_c) sum_m |psi_m>rho_c -> spectrum, subsystems, mode coherence, row scores
reported result = comparison with matched controls
```

Matched encodings

The product control and the coupled map have the same rows, angle map, and two local half-layers. Only the interaction changes. At J = 0 the two RY(x/2) layers compose to the ordinary real product RY(x) encoding.

Encoding	Definition	Purpose
J = 0	RY(x/2) - I - RY(x/2)	Real product-angle baseline.
Partial J	RY(x/2) - ZZ(J) - RY(x/2)	Adds deterministic conditional-dependence interactions.

How J is obtained

```
Sigma = covariance of standardised analysis rows
Theta = inverse(Sigma + epsilon I)
J_jk = -s Theta_jk / sqrt(Theta_jj Theta_kk), j != k
J_jj = 0
```

- J is not learned from accuracy or a predictive loss.
- There is no dataset-wise rescaling that forces max |J_jk| = 1.
- The scale is declared before an audit: s = 1.0 for Bowles and s = 0.8 for topology, Iris, and fraud.
- Every held-out audit fits scaling, J, rho, and thresholds without using held-out labels.

Simple reading

We deliberately give the Hilbert-space representation a fair chance to add structure, then ask whether that structure is stable and relevant. A nonzero quantum diagnostic is not automatically a useful data result.

2. What the reported numbers mean

Diagonalising rho gives nonnegative weights lambda_0 >= lambda_1 >= ... that sum to one. These weights describe how concentrated the encoded empirical mixture is. Subsystem and mode-coherence readings ask different questions and must not be collapsed into a single score.

Reading	Calculation	Plain interpretation
Purity	sum_i lambda_i^2	Larger means the encoded class mixture is concentrated into fewer directions.
Effective rank	1 / purity	Approximate number of substantially occupied spectral directions.
Mass gap	log(lambda_0 / lambda_1)	Separation between the leading direction and its nearest competitor.
Third moment	Tr(rho^3)	Another concentration summary, more sensitive to dominant eigenvalues.
Mutual information I	Mean single-mode/rest entropy balance	Total dependence across one encoded mode versus the remaining modes.
Log negativity Q	Mean partial-transpose trace norm	Entanglement witness across single-mode cuts; Q > 0 certifies representation entanglement.
Encoded current	max |Im Gamma_jk|	Basis-explicit antisymmetric mode coherence. It is not an entanglement witness.
Row fidelity Fs		How much of a row lies in the leading subspace learned for class c.
Anomaly score	1 - Fs	Large values mean a row lies outside the leading normal-reference subspace.
Bargmann phase	arg product of closed-loop overlaps	Gauge-invariant geometric phase of a chosen loop; not automatically topological.

Why the ordinary quantum kernel is not enough

Let A contain the statevectors as columns and G = A^dagger A be the complex Gram matrix. The nonzero spectrum of rho = A A^dagger / M equals the spectrum of G/M. Pairwise overlaps therefore retain the operator spectrum. Information is discarded when the standard fidelity kernel replaces G_mn by |G_mn|^2, because overlap phases and closed-loop phase products disappear.

Interpretation rule
Purity, Q, or current can show that an encoding created operator structure. Relevance requires another control: stability, a null test, or a held-out data question. The report always states both layers.

3. Where each computation is used

Data	Size / features	Fit and control	Question
Synthetic exact checks	Small 3-qubit tables	Same rows under J = 0 and coupled maps	Does the code obey the algebraic identities and phase conventions?
Ring vs disk	300 + 300 rows; 3 features	Means and covariance matched to machine precision; 500 permutations	Do current summaries reveal a known support hole beyond first two moments?
Bowles families	55 variants; 2-16 native features	Native classical EDA plus matched J = 0 / J; no PCA	Does qEDA provide an ordering distinct from selected covariance summaries?
Iris	150 rows; 4 native features	Five-fold cross-fitting; three classical error probes	Are hard held-out rows also less compatible with the true-class operator?
Credit-card fraud	284,807 rows; 30 released variables	Normal-only fit and calibration; natural-prevalence test	Can a normal-reference operator score anomalies, and does coupling add information?

High-dimensional handling in the fraud audit

Thirty variables do not fit the declared four-qubit full-operator budget. No XGBoost ranking, predictive importance, PCA, or first-column rule is used. Four seeded permutations and all 30 cyclic shifts create 120 paths. Each path uses four variables, and every variable appears exactly 16 times. Names are sorted before wire assignment, so stored column order does not choose the result.

```
30 variables x 4 cyclic base paths = 120 four-variable marginals
each variable occurs 4 x 4 = 16 times
path score -> aggregate by median (primary) or 90th percentile (sensitivity)
```

Fraud split

- A stratified 80/20 split gives 56,962 held-out transactions, including 98 frauds.
- 5,000 normal training rows fit scaling, J, rho, and the leading projector.
- A disjoint 10,000 normal rows calibrate target false-positive rates.
- Fraud labels are used only for held-out evaluation.

Important data limitation

The released variables V1-V28 were already anonymised by PCA before publication. We apply no additional PCA. The conclusions concern the released representation, not the unavailable original transaction fields.

4. E1 - algebraic and implementation controls

These checks are not data discoveries. They verify that later differences are produced by the declared map rather than a sign, normalisation, or PennyLane convention error. The vectorised fraud encoder is additionally checked state by state against PennyLane; its maximum amplitude error is 2.22e-16.

Experiment	Metric	J = 0	Partial J	Meaning
operator gram	spectrum max abs error	1.94e-16	1.11e-16	Nonzero spectra of rho and G divided by M
operator gram	gram max abs imaginary	0	0.04408	Complex overlap structure appears after coupling
operator gram	bargmann phase radians	0	0.00456	Three-state closed-loop phase
synthetic profile	purity	0.5958	0.7124	Same correlated synthetic table
synthetic profile	mutual information bits	0.04117	0.4499	Mean single-mode versus rest
synthetic profile	log negativity	0	0.4066	Mean single-mode cut
synthetic profile	max encoded current	0	0.06436	Basis-explicit imaginary mode coherence
current witness	max encoded current	0	0.06876	Independent real-control check

What this establishes

- The class-operator spectrum and scaled complex Gram spectrum agree to machine precision.
- The real J = 0 control has no imaginary overlap structure and zero encoded current in the declared basis.
- The coupled map can create overlap phase, entanglement, mutual information, and current.
- The last statement concerns the representation. It does not yet say that the added structure answers a data question.

E1 conclusion

The implementation passes the required identities. We can therefore interpret later negative results as genuine audit outcomes, not as a failure to generate quantum operator structure.

5. E2 - matched-moment topology audit

The ring and filled disk have visibly different supports, but separate invertible recolouring makes their empirical means and covariance matrices equal to 1.49e-16 and 2.00e-15. The test asks whether the current operator summaries recover the hole after this classical second-order information has been matched.

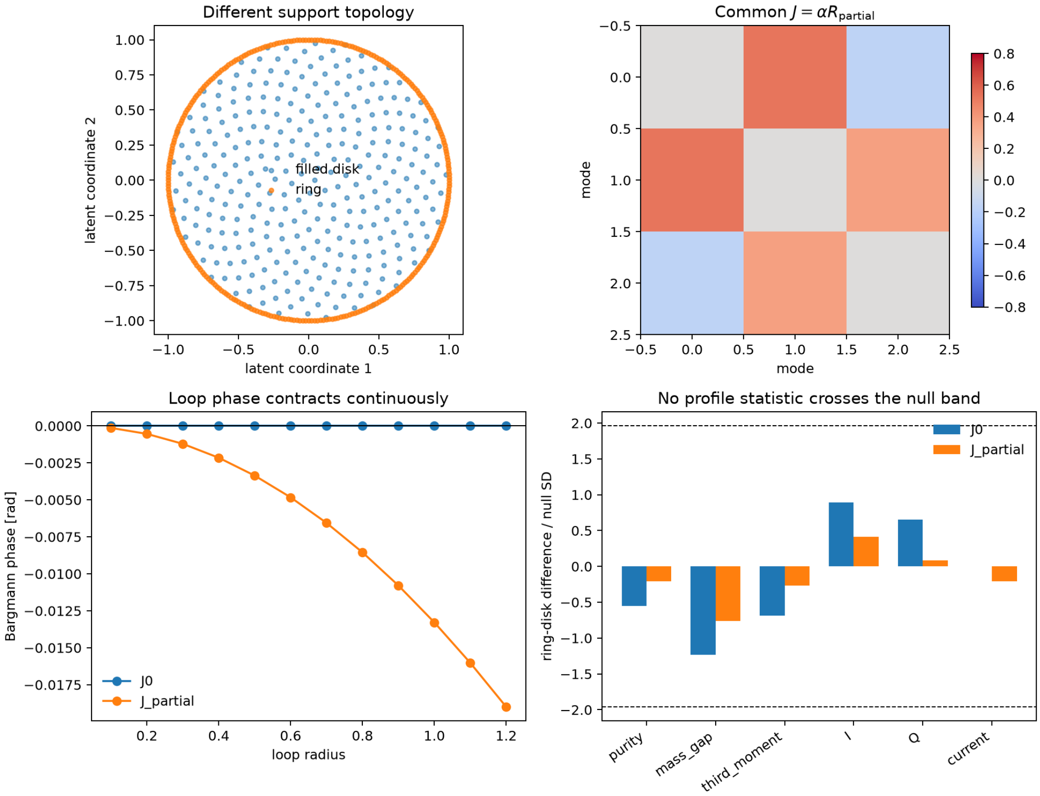

Figure 1. The clouds have matched first two moments. Coupling creates a geometric loop phase and nonzero operator structure, but the permutation statistics remain inside their null bands.

Metric	p, J = 0	p, partial J
purity	0.575	0.840
mass gap	0.224	0.447
third moment	0.483	0.808
I	0.397	0.667
Q	0.585	0.944
current	1.000	0.858

The coupled operators are genuinely enriched: mean Q is 0.387 and mean maximum current is 0.038. Nevertheless, every two-sided permutation p-value exceeds 0.22. The coupled Bargmann phase changes continuously from -1.35e-04 to -1.90e-02 radians as the loop expands. It tends to zero when the loop contracts.

E2 conclusion: a useful negative result
The encoding creates entanglement, current, and geometric holonomy, but none of the reported statistics detects the support hole. Geometric phase is not a protected topological invariant here. More quantum structure does not imply more relevant data information.

6. E3 - Bowles benchmark families

The original benchmark generators and random streams are replayed, but the generated tables - not the published models - are the objects of study. Standard EDA is calculated on all native features. The correlation spectrum is inspected but no principal-component scores are formed and no directions are removed.

All 55 variants receive standard EDA. Full class operators are computed for 54 variants, producing 108 binary-class operators on 2-10 qubits. The 4 x 4 Bars-and-Stripes table retains all 16 features and is explicitly marked outside the full-operator budget.

Family median	Mean |r|	Corr. rank/d	Fisher	Delta purity	Delta gap
Bars and Stripes	0.172	0.509	0.001	-	-
Hidden manifold	0.311	0.585	0.123	0.156	0.440
Hidden manifold (difficulty)	0.303	0.454	0.133	0.134	0.358
Hyperplanes parity	0.804	0.142	0.237	0.173	0.577
Linearly separable	0.051	0.973	0.144	0.289	1.093
Two curves	0.807	0.221	0.046	0.150	0.262
Two curves (difficulty)	0.822	0.140	0.045	0.158	0.178

How to read the table

- Mean |r| and correlation rank/d are native-feature summaries.
- Delta means partial-J minus the matched J = 0 operator on the same rows.
- Positive Delta purity and Delta gap mean the coupled operator concentrates weight and separates its leading mode.

The linearly separable family has the weakest median marginal correlation (0.051) and an almost full native correlation rank (0.973), yet it has the largest median spectral sharpening (Delta purity 0.289; Delta gap 1.093). Across the 54 profiled variants, Spearman correlation between mean |r| and Delta purity is -0.234, and with Delta gap it is -0.420.

E3 conclusion
For these declared summaries, qEDA is not merely renaming mean correlation or Fisher separation. This is relative empirical nonredundancy, not a claim that no classical statistic could reproduce the same ordering.

7. E4 - held-out real-data audits

Iris: sample-level compatibility

All four native Iris features fit the full operator. In each of five stratified folds, every transformation and class operator is fitted on four folds. The held-out row is scored against the leading subspace of its true-class operator. A separate union of logistic-regression, LDA, and RBF SVM errors marks difficult rows.

Rows	Probe errors	Q	Current	Fs correct	Fs wrong	p
150	7	0.420	0.047	0.933	0.823	4.91e-04

The seven rows missed by at least one classical probe have lower true-class fidelity: 0.823 versus 0.933. This supports a sample-level reading on one small dataset; it is not a general theorem about classifier errors.

Credit-card fraud: normal-reference anomaly scoring

Method	AUC	AP	Observed FPR	Recall	Precision
qEDA J = 0 median	0.961	0.274	0.0051	0.704	0.192
qEDA partial J median	0.960	0.277	0.0051	0.724	0.196
Isolation Forest	0.951	0.128	0.0050	0.429	0.128

The table uses the median of 120 marginal anomaly scores and a target FPR of 0.005 calibrated on separate normal training rows. The test set retains the natural fraud prevalence. Isolation Forest is context only and does not define qEDA.

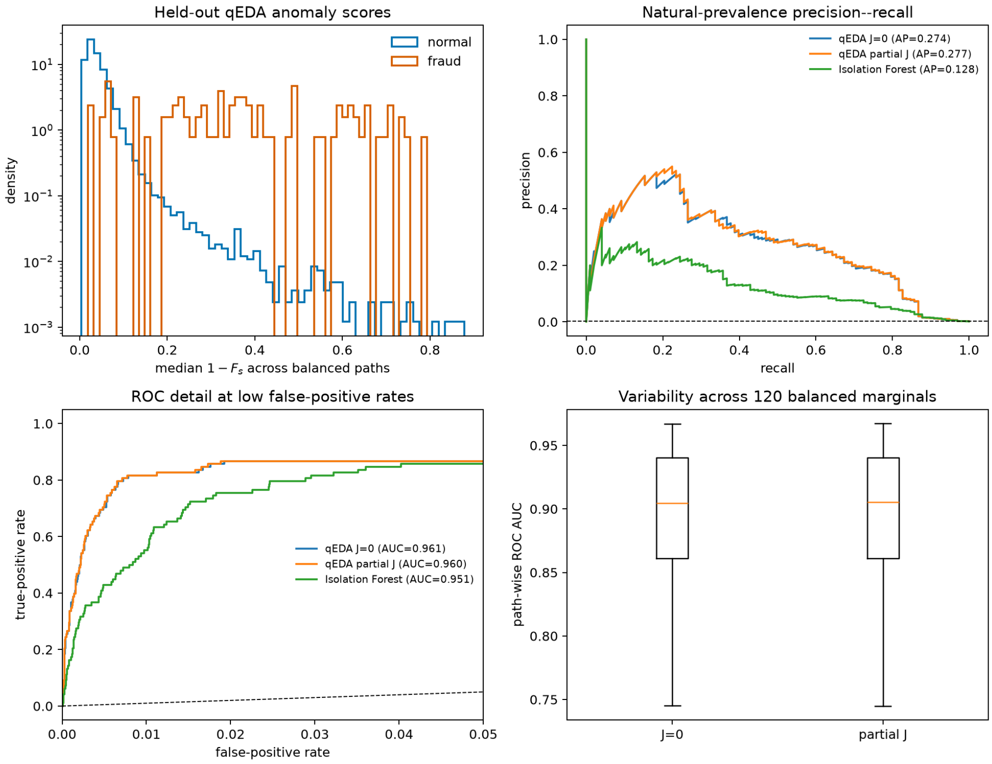

Figure 2. Both qEDA profiles rank many frauds above normal transactions. The J = 0 and partial-J curves nearly coincide despite nonzero coupled negativity and current.

8. What the fraud result actually says

The partial-J representation is not trivial. Across the 120 paths, median Q is 0.0566 and median current is 0.0242; both are zero for the real control. Yet the added structure does not materially change anomaly ranking.

Path-level comparison	Median value	Reading
Delta purity	4.428e-04	Almost no spectral concentration change.
Delta AUC	-7.985e-06	No systematic ranking improvement.
Delta average precision	6.904e-06	No systematic precision-recall improvement.
Coupling Frobenius norm	0.1416	The coupling is present but modest.
Maximum |J_jk|	0.0769	Weak dependence is not inflated to one.

Audit decision

Keep the Hilbert-space anomaly score if it is useful, but the current released fraud representation provides no empirical reason to pay for the coupled layer. The simpler J = 0 encoding is the defensible choice for this task unless a different, predeclared data question shows a stable gain.

Reproduction map

Command	Primary outputs
python step1_statistics.py	Console algebraic checks.
python topology_holonomy_test.py	output/data/topology_*.csv and topology figure.
python benchmark/spectral_benchmark.py ...	output/benchmark classical, spectral, and coverage tables.
python e4_model_errors.py	Cross-fitted Iris console summary.
python fraud_qeda_case_study.py	Fraud path, score, summary tables, and figure.

Scope boundary

- All computations are classical simulations of declared Hilbert-space embeddings.
- Results are conditional on preprocessing, encoding, scale, operator readings, and controls.
- Nonzero negativity or current certifies representation structure, not usefulness or topology.
- The current paper is an operational guide for analysts and QML researchers. QIFT carries the deeper field-theoretic framework.

Bottom line

qEDA is valuable even when it recommends against a more elaborate encoding. Its role is to make the cost and content of a Hilbert-space representation visible before that representation is hidden inside a model.

Data provenance: synthetic benchmark generators follow Bowles, Ahmed, and Schuld, Better than classical? The subtle art of benchmarking quantum machine learning models (arXiv:2403.07059). Iris is supplied by scikit-learn. The fraud table is the public anonymised ULB/Kaggle release.